

KELAN LIU

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WORK EXPERIENCE

CLASSLINK

Software Engineer

Jan 2024 – Present

- Engineer on the Core Services team owning **10+ foundational APIs/services** powering **SSO** and app launches for **25M+** students, teachers, and admins daily (Applications APIs, SAML Console, Management Console, AppTrack, ThreatScan)
- Migrated core API services from **EC2** to **ECS Fargate**, standardizing deployments and **reducing peak-login p99 latency by 30%**
- Cut critical query costs up to **68%** and built a resilient retry library (exponential backoff) that eliminated socket hangups, improving reliability of user-facing flows
- Designed and shipped a serverless translation pipeline (**Lambda/S3/AWS Bedrock**) enabling localization across multiple product front ends; presented architecture department-wide
- Partnered **cross-functionally** with the Auth team to ship Microsoft Entra and Google Directory integrations, contributing directly to adjacent team codebases

WAYMO

Software Engineer, Machine Learning, Research Intern

Sept 2021 – Jan 2022

- Contributed to a real-time autonomous driving **simulation** platform through kinematics modeling and cloud visualization integration (**C++**, **Python**)
- Partnered with Simulation and UI teams to refine scenario realism and optimize user interaction workflows, improving fidelity across kinematics modeling, visualization, and driving UI
- Built an operator-facing driving UI and a **data collection pipeline** for simulation workflows (**Angular**, **RxJS**, **TypeScript**, **Protobuf**)
- Optimized end-to-end latency to under **200ms** by implementing **D3.js**-based visualization tools for monitoring rendering and mapping plugin performance

INTUIDEX

Software Engineer Intern

May 2021 – Aug 2021

- Ported and optimized **video-processing** and **ML** algorithms for embedded **GPU** systems using **CUDA**, **Docker**, and **Python**
- Implemented and optimized real-time motion detection algorithms supporting a **YOLO**-based license plate detection pipeline for the **Jetson Nano** and **Xavier (OpenCV)**
- Cut YOLO training time by **30%** by improving upstream data quality, enabling faster model convergence

PROJECTS

[Creative Coffee Club](#) | *Next.js 15, React 19, Supabase, Stripe, Twilio, Telnyx, SendGrid, PostHog, Vercel*

- Built and shipped a social platform for **~850** active creatives, scaling **MRR** to **\$26K** via **Stripe** subscriptions with **A/B tested** price points, plus an automated **Twilio SMS** pipeline handling availability polling, group introductions, reminders, and delivery tracking
- Developed a scoring-based matching algorithm with city/slot segmentation and preference weighting, plus a full admin suite with dry run previews, manual overrides, CSV exports, and real-time SMS logs
- Integrated **PostHog** analytics and experiment tracking, **SendGrid** drip campaigns, **Supabase** auth with magic links, and 7 automated cron jobs on **Vercel** powering the end-to-end biweekly matching cycle

GramIt (AWS Game Builder Challenge) | *AWS Serverless, React Native, XState, TypeScript, Python*

- Developed GramIt, an online multiplayer game combining social interaction and creative image selection (**React Native**, **AWS**)
- Backend built with **AWS Step Functions**, **Lambda**, and **API Gateway WebSockets** for real-time game state management
- Integrated **XState** for frontend state management, aligning it with AWS Step Functions to ensure seamless game flow
- Deployed the game on **AWS Amplify** and utilized **Amazon Q** for **optimizing** Step Function orchestration, **enhancing** system efficiency

EDUCATION

UNIVERSITY OF TEXAS AT AUSTIN

Master of Science in Computer Science, Concentration in Machine Learning & Artificial Intelligence

LEHIGH UNIVERSITY

Bachelor of Science in Computer Science, Minor in Japanese Language and Literature

Coursework: Data Structures, Software Engineering, Algorithms, AI, Databases, Systems, Computer Architecture, System Design, OS

TECHNICAL SKILLS

- **Languages:** Python, TypeScript/JavaScript, SQL, HTML, CSS
- **Tools:** Docker, DevOps, Git, Claude Code, Jira, REST, GCP, AWS, PostgreSQL, MySQL, Redis, gRPC, Protobuf, Linux, Postman
- **Frameworks/Libraries:** React, Angular, Flask, Node, RxJS, D3.js, NumPy, Pandas, Matplotlib, Django, PyTorch, OpenCV, SK-learn
- **Certifications:** AWS Certified Developer